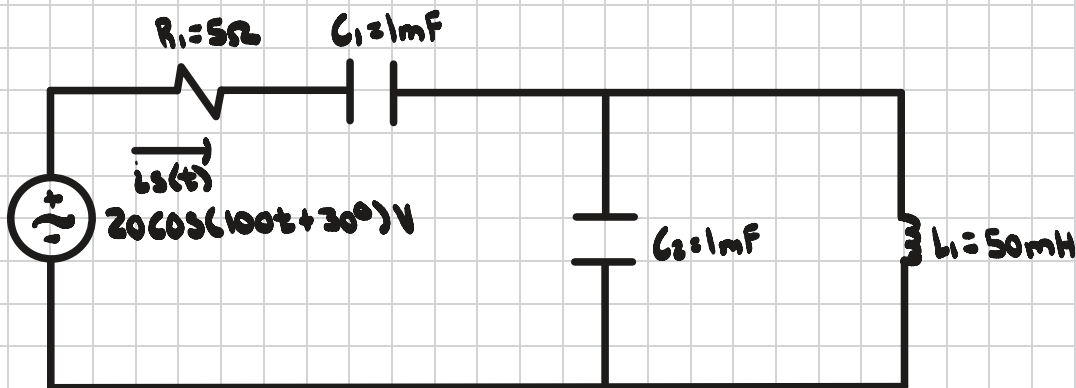
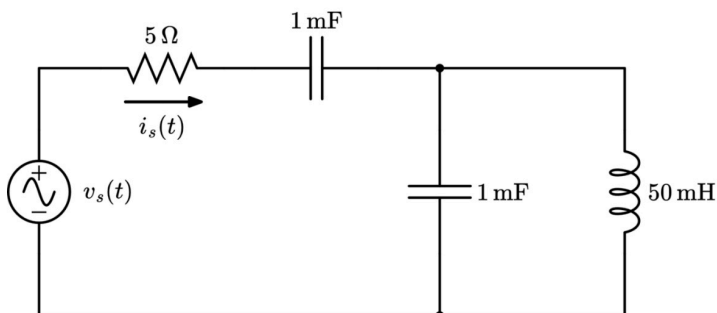


For the circuit shown below, find $i_s(t)$.

Suggested time limit: 10 minutes

$$v_s(t) = 20 \cos(100t + 30^\circ) \text{ V}$$

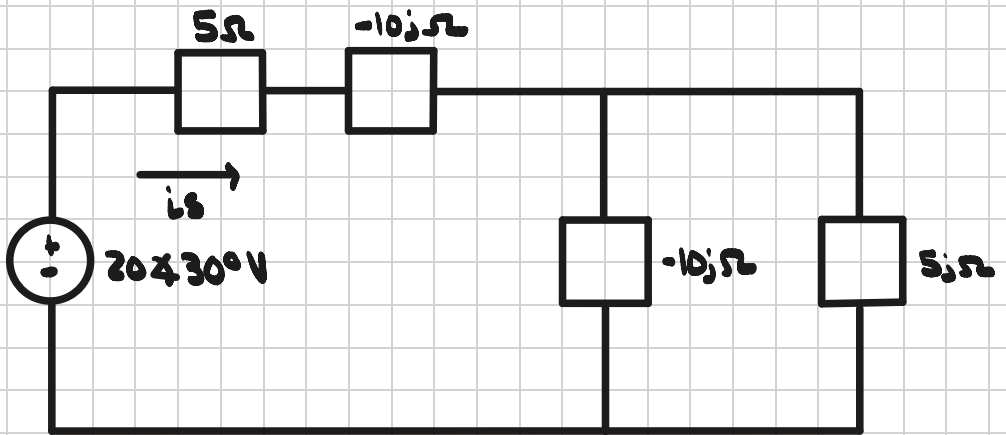


$$V_s = 20 \angle 30^\circ \text{ V}$$

$$Z_R = R_1 = 5\ \Omega$$

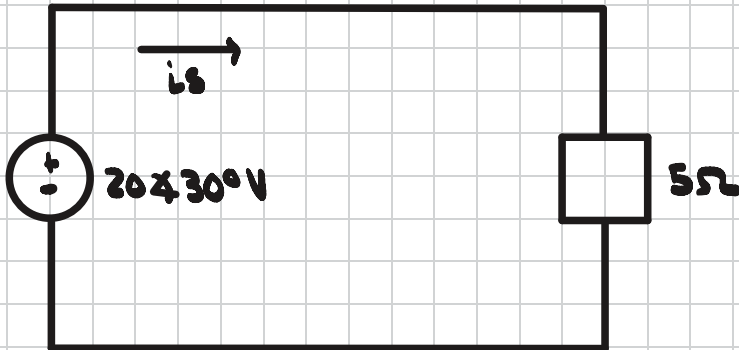
$$Z_{L1} = j\omega L = j(100)(50 \cdot 10^{-3}) = 5j\ \Omega$$

$$Z_{C1} = Z_{C2} = \frac{1}{j\omega C} = \frac{1}{j(100)(1 \cdot 10^{-3})} = -10j\ \Omega$$



$$Z_{eq} = (-10j \parallel 5j) + 5 - 10j$$

$$Z_{eq} = 5\Omega$$



$$i_s = \frac{20\angle 30^\circ}{5} = 2\sqrt{3} + 2j \text{ A} = 4\angle 30^\circ$$

$$i_s(t) = 4\cos(100t + 30^\circ) \text{ A}$$